

Siddhi Singh Rajput

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EDUCATION

Indian Institute of Information Technology Senapati, Manipur

Electronics and Communication Engineering - VLSI

Imphal, Manipur

2025 – Present

Gyansthal Public School

Higher Secondary, Science

Mahoba, Uttar Pradesh

March 2024

PROJECTS

Air Draw

Python, FastAPI, OpenCV, MediaPipe, WebSockets, HTML5 Canvas, JavaScript, Tesseract OCR

- Built a real-time hand tracking system using OpenCV and MediaPipe to detect **21 hand landmarks** for gesture-based drawing, erasing, and drag operations.
- Implemented asynchronous FastAPI WebSocket communication for low-latency MJPEG streaming and real-time stroke synchronization between backend and frontend.
- Integrated Tesseract OCR and contour approximation algorithms to recognize handwritten air text and auto-correct freehand shapes into structured geometry.

GitHub Profile Analyzer

React, Vite, Tailwind CSS, Axios, React Router, Recharts, GitHub API

- Built a React-based web application to analyze public GitHub profiles and repository activity using GitHub REST APIs.
- Implemented modular API services, custom React hooks, and parallel data fetching to improve scalability and reduce response wait times.
- Visualized programming language distribution and repository statistics using interactive Recharts-based pie charts and responsive dashboards.
- Integrated error handling, dynamic routing, and responsive Tailwind UI components for improved frontend reliability and user experience.

Indoor Telemetry Dashboard — Hackathon

Kotlin, Android BLE, React, Supabase, OkHttp, JavaScript

- Developed a real-time indoor positioning system using BLE beacons and IMU sensor data for stable coordinate tracking inside a 10×10m environment.
- Implemented sensor fusion using BLE distance weighting and dead reckoning techniques to improve positioning consistency and movement tracking accuracy.
- Built a React dashboard with live telemetry visualization, anchor distance monitoring, and realtime Supabase subscriptions for synchronized device updates.
- Integrated Supabase PostgreSQL backend with REST-based mobile writes and realtime frontend communication for low-latency telemetry updates.

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript

Libraries & Tools: OpenCV, MediaPipe, FastAPI, Git, GitHub, VS Code

Web Technologies: HTML5 Canvas, WebSockets, React.js

Databases: MongoDB

Data & Analytics: Power BI, MS Excel

LEADERSHIP & RESPONSIBILITIES

Contributor, Girls Script Summer of Code 2026

- Contributed to collaborative open-source projects using GitHub workflows and team-based development practices.
- Worked on issue-based development tasks and improved code collaboration through pull request reviews.

Contributor & Campus Advocate, Social Summer of Code 2026

- Participated in community-driven development programs and contributed to real-world coding projects.
- Promoted technical learning initiatives and student engagement through campus-level community activities.

Google Gemini Campus Ambassador

- Encouraged AI awareness and peer collaboration through technical discussions and community outreach programs.
- Connected students with AI-focused learning resources, workshops, and technical communities.